Track Stack: Complete Product & Feature Walkthrough

Detailed guide to platform motive, user roles, core principles, and end-to-end walkthrough of every single screen.

1. Purpose, Motive & Institutional Problem

Modern college extracurriculars are plagued by disorganized administration. Student attendance is collected via paper sheets or Google Forms prone to proxy signing. Assignment deliverables are shared over WhatsApp groups where feedback is lost. Certificates are easily forged, and graduating students leave university without a unified, verified transcript of their co-curricular achievements.

Track Stack solves this problem for Mirai School of Technology (MSOT) by delivering a unified, verified, and low-effort ecosystem built on three unshakeable pillars:

* VERIFIED: Every attendance mark, submission, evaluation score, and certificate is backed by cryptographic proofs, timestamped logs, and immutable audit trails.
* LOW EFFORT FOR COORDINATORS: Student coordinators can manage an entire Saturday Club Day session directly from a mobile phone in seconds using single-tap controls.
* STUDENT-OWNED RECORD: Every student can download a tamper-proof Co-Curricular Transcript PDF with embedded QR verification, ready for placement drives and higher studies.

2. Role-Based Governance Model

Track Stack provides specialized operational dashboards and workflows for five user roles:

* System Administrator (Ayush Chauhan): Global governance, institution-wide settings (max 2 clubs per student, grace periods), immutable audit log inspection, and Lumi AI knowledge-base management.
* Campus Manager (Sundaram Mishra): Delhi-NCR campus oversight, Duty Leave review and approval for inter-campus competitions, and review of mathematically reconciled monthly reports.
* Faculty Advisor (Gaurav Kumar): Reviews and approves proposed Saturday Club Day plans, conducts 0-10 rubric evaluations, conceals draft scores, and issues achievement certificates.
* Student Coordinator (Rishav Raj): Controls the live mobile Run Deck for their club, projects the rotating QR code, marks rosters with single-tap controls, corrects attendance with mandatory reasons, and posts club announcements.
* Student (Chintu): Explores clubs, enrolls under the 2-club limit, scans rotating QR codes, submits project deliverables with revision versioning, registers for hackathons, and downloads verified certificates and transcripts.

3. Detailed Walkthrough of Every Single Feature

A. Rotating QR Anti-Proxy Check-In

The coordinator screen displays a live QR code that regenerates every 25 seconds with an animated countdown ring and a large tabular 6-digit OTP. The server verifies time-stepped HMAC-SHA256 signatures, rejecting forwarded screenshots. Students inside the 300m campus geofence check in instantly.

B. Submissions Engine & Revision History

Students upload deliverables in PDF, PPTX, DOCX, ZIP, or external repository links. When resubmitted before the deadline, the platform increments the version counter (v1, v2) rather than overwriting files, preserving the complete academic history for faculty review.

C. Rubric Evaluation with Draft Score Concealment

Evaluators grade submissions on 0-10 rubrics across Technical Depth (40%), Presentation (30%), and Participation (30%). Scores remain strictly concealed as 'Draft' until officially published to maintain fair grading across all students.

D. Cryptographically Verified Certificates

Certificates feature an embedded cryptographic HMAC signature and scannable QR code. Entering any certificate ID on the public lookup page (/verify/[id]) recomputes the signature against database records, proving authenticity (VALID) or flagging modifications (INVALID-TAMPERED).

E. Automated Inter-Campus Duty Leave

When a student registers for an approved inter-campus competition (such as Gambade or HackTech), Track Stack automatically generates a Duty Leave request. The Campus Manager approves it with comments, producing an official signed PDF letter for faculty clearance.

F. Co-Curricular Transcript

Students can download their complete university transcript anytime. It synthesizes club memberships, attendance percentages, rubric scores, and earned certificates into a vector PDF with a verification QR code, backed by an authorized /api/transcript/[studentId] API endpoint.

G. Reconciled Monthly Reports & Scheduled Cron Flow

At month-end, automated jobs compile comprehensive club performance summaries sent to Campus Managers. Report figures strictly match raw transactional database records (0.00% discrepancy), downloadable in vector PDF and CSV formats.

H. Meet Lumi: The Campus AI Assistant

Lumi is Track Stack's official conversational companion, available via a floating glowing avatar drawer on every screen. Powered by Google Gemini Flash and a 60+ FAQ scored knowledge base, Lumi answers questions in English, Hindi, and Hinglish. It securely calls read-only tools to check attendance, club status, and deadlines, while maintaining firm guardrails against jailbreaks and unauthorized grade edits.

4. Clubs Covered on Track Stack

Track Stack governs 4 core student clubs and Kaizen Society's 3 specialized sub-clubs:

* NexCell (Entrepreneurship Club) - Startup incubation, venture pitches (#3B9BE0)

* Malhar (Cultural Club) - Music, dance, dramatics, creative visual arts (#E8A935)

* Khel Society (Sports Club) - Athletics, football, inter-college tournaments (#1E6FC0)

* Kaizen Society (Technical Club) - Flagship engineering club with 3 chapters:

- Dev Club (Kaizen Dev Club) - Full-stack, mobile, open source (#4FA39A)

- CP Club - Competitive coding, algorithms, LeetCode/Codeforces (#A3E635)

- KI (Kaizen Intelligence) - Machine learning, computer vision, AI (#D9A441)